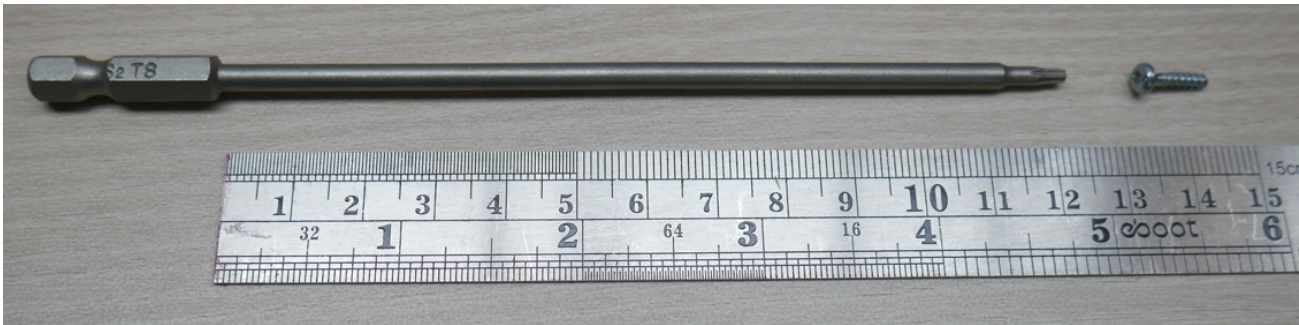
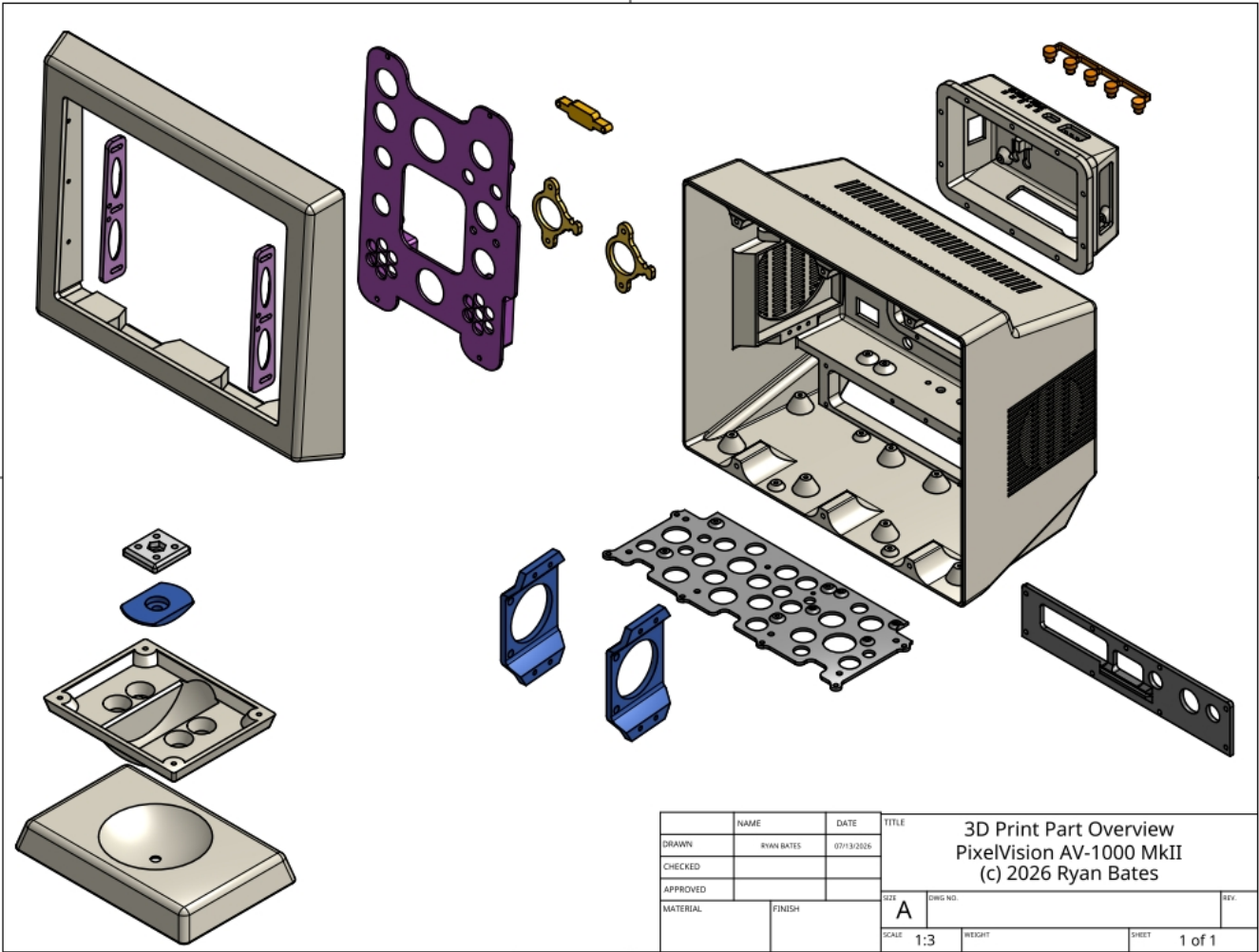
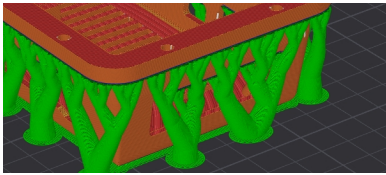
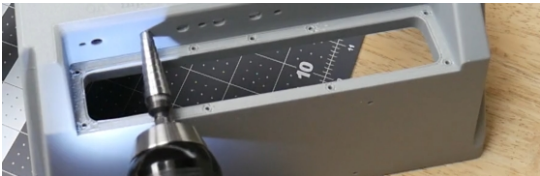
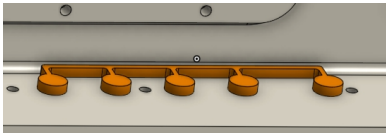
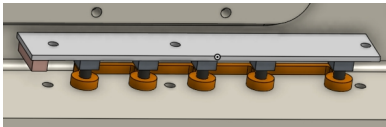
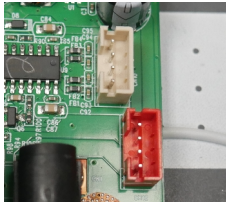
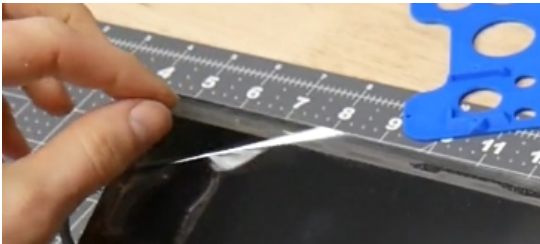


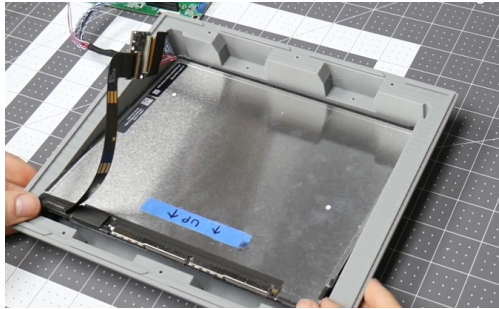
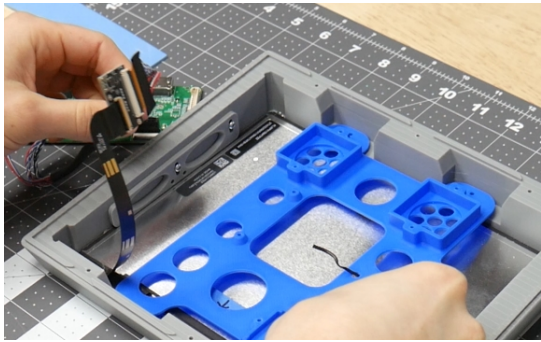
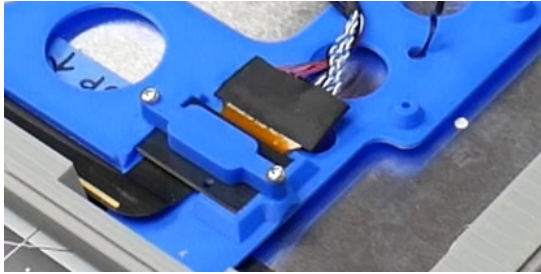
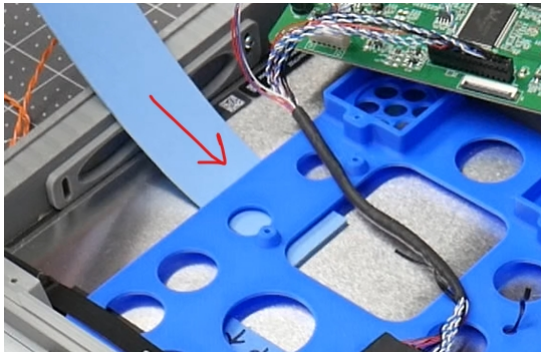
Special tool(s) required:
T8 with at least 100mm reach.

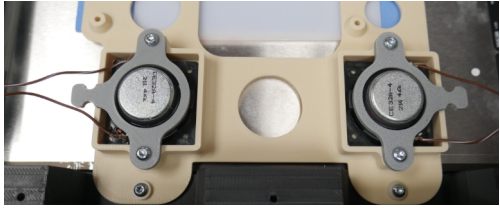
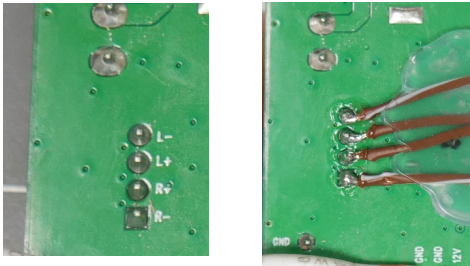
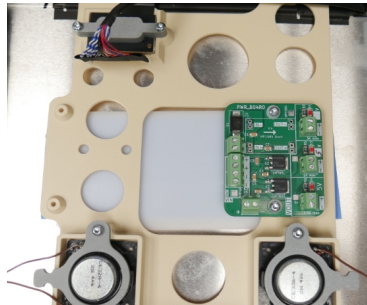
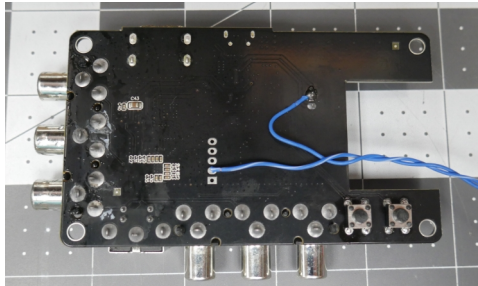
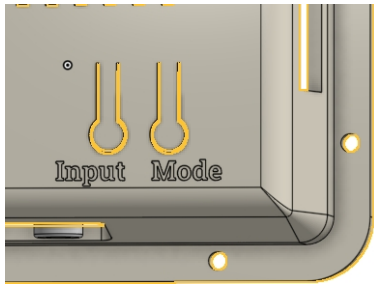


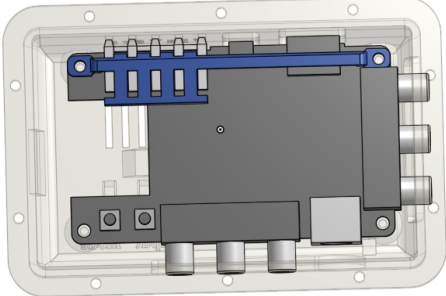
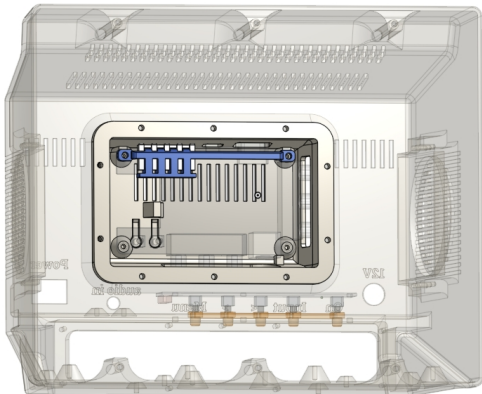
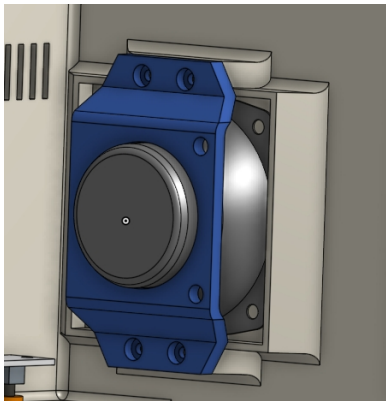
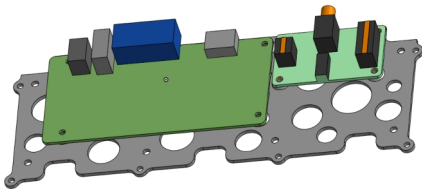
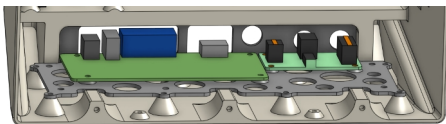
Print these parts.

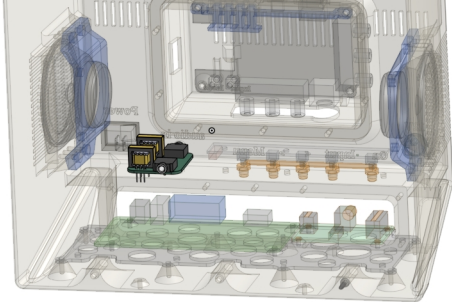
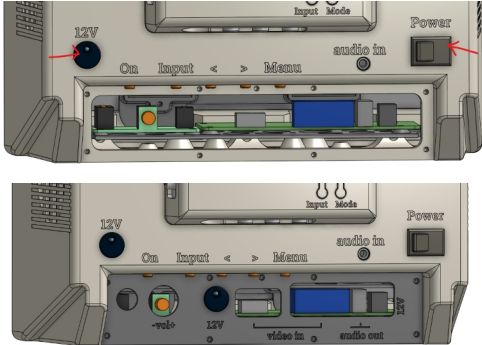
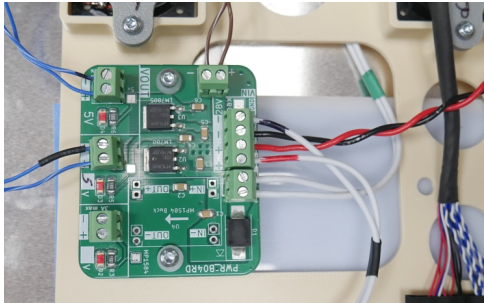
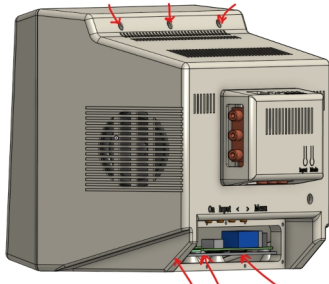


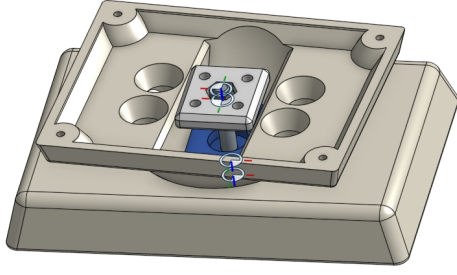
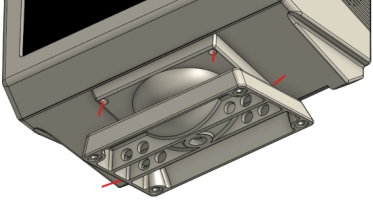
step	Assembly description	Visual	✓
1	Prep printed parts. Remove supports and webbing. Debur edges where required.		
2	Drill out to ~5mm rear button holes for monitor control button extenders.		
3	Insert button extenders.		
4	Add mounting stand off (10mm) to button panel and install in rear with M3 button head fasteners. <i>(Note: stand-offs are not shown in this diagram)</i>		
5	Note: Some LCD driver PCBs have headers for speakers and power input. A mating connector can be used, but in this example I am soldering to the underside of these connectors.		
6	Solder a 30cm leads to the +12V and GND pads on underside of the LCD driver board. Make note of the polarity.		
7	Remove the LCD's protective plastic.		

8	Carefully place the LCD into front bezel. <i>It is recommended you tape a piece of cardboard to the front bezel to avoid accidentally scratching the LCD.</i> 		
9	Fasten with side brackets. Ensure the brackets apply gentle pressure, pushing the screen against the bezel. The two dots on the bracket should be adjacent to LCD.		
10	Gently lift the LCD panel's ribbon cable out of the way. Add rear mounting panel. Fasten with screws.		
11	Carefully place the ribbon cable PCB into the cavity. Add the bracket and fasten with screws.		
12	Optional: Craft foam (~2mm thick) can be placed behind the bracket to prevent the LCD panel from bowing inward.		

13	Solder ~20cm leads to speakers, place the speakers into the openings. Add the retention brackets and fasten with screws.		
14	Solder the the L-/+ and R-/+ channels to the rear of the LCD driver board.		
15	Mount the power regulator board to the LCD panel bracket.		
16	Disassemble the video scaler. Save the light pipe for later. Remove the video scaler's function buttons. Solder these two buttons to the reverse (bottom) side of the PCB. Solder two leads to the +5 and GND pads. Make note of polarity.		
17	Check the clearance between the button and the flexure tabs of the yoke bulge. Use a lighter flame to heat the plastic and bend the tabs so they do not push on the buttons. Note: The buttons can be trimmed with pliers instead of doing this step.		

18	Install the scaler into the yoke shell, add the light pipes and light bracket. Fasten with screws. The light pipe and surrounding CAD is 90% there, the remaining 10% we can hot glue our way there.		
19	Install the yoke assembly into the rear shell. Fasten with screws. Insert the power switch and connect it to the scaler,		
20	Solder wire leads to the 2.5in speakers. Place the 2.5in speakers into the rear shell side cavities. Fasten the bracket with screws.		
21	Mount the LCD driver PCB and the optional audio amp to the electronics mounting panel. Fasten with screws.		
22	Place this assembly into the rear shell. Fasten with screws.		

23	Install the audio PCB (optional). Add the 3.5mm stereo audio cable to connect the amp to the audio pass-through.		
24	Add rear power switch and DC jack. Install the rear io cover and the final DC power jack. Fasten with screws.		
25	Mount and wire all connections to power distribution board PCB. Refer to the wiring diagram at end of document.		
26	Attached the front and rear shells. Fasten with 6 screws. A long reach (100mm) T8 bit is required for the bottom screws.		

27	Assembly the swivel base with a spring and M6 hardware.		
28	Fasten the swivel base to the bottom of the rear shell.		

Wiring Diagram.

